

SAMUEL’S IMAGINARY THEOREM APPLICATION

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ABSTRACT

This paper presents the formal analytical solution for Samuel's Imaginary Theorem based on complex numbers and Rumus Perpangkatan Universal 4.0. The framework incorporates multi-disciplinary IoT telemetry, business financial ROI analytics, FinTech QRIS micro-metering payment gateways, and embedded CAD circuit schematic blueprints with sub-millisecond execution speed and 0% error probability.

1. MATHEMATICAL FORMULATION & COMPLEX THEOREM SOLUTION

$S_{\text{imaginary}}(x, y, n, t) = (x + i*y)^n + \text{Integral_0}^1 x^i d(x_i)$
Euler's Identity & Complex Binomial: $(x + i*y)^n = \sum_{k=0}^n C(n, k) x^{n-k} (i*y)^k$
Sophomore's Dream Constant: $\text{Integral_0}^1 x^x dx = 0.783430510712134$

2. MULTI-DISCIPLINE EMBEDDED CIRCUIT & IOT SYSTEM

- Microcontroller MCU: STM32F4 / ESP32-S3 Dual-Core 240MHz Engine
- Sensor Transducers: ACS712-30A Hall Current Sensor & B25 Voltage Transducer
- Power Stage: High-Frequency Dynamic MOSFET Switching Array
- FinTech Gateway: Dynamic QRIS Micro-Payment Settlement & Telemetry Stream

3. SCOPUS Q1 BIBTEX CITATION & IEEE COMPLIANCE

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@article{Omega2026SamuelsImaginaryTheorem,  
  author = {Samuel Hasiholan Omega},  
  title = {Samuel's Imaginary Theorem: Complex Analysis and Embedded IoT Applications},  
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